Exercises

In some of the following questions, an object resting on an incline, as shown on the right side of Figure 7.1. The object experiences Earth's gravitational force, which acts vertically from the center of the object toward the Earth's center. Assuming the triangle is a hard object, the forces can be resolved into two components: (i) the force dragging the block down the slope (labeled as $F2$) and (ii) the force acting perpendicular to the slope known as the normal force (labeled as $F1$). If the gravitational force is given by mg (where m is the object's mass and g is the gravitational constant), what the magnitude of $F1$ and $F2$ are as follows:

$$F1 = mg\cos(\theta) \quad F2 = mg\sin(\theta)$$

We can also write Newtons second law using three ordinary differential equations (ODE) as follows:

$$\text{(i) } F = m\frac{dv}{dt} \quad \text{(ii) } F = mv\frac{dv}{dx}$$

All are equal but allow you to calculate speed as a function of time OR speed as a function of distance.

Question 7.1: An object, mass m, is at rest on a plane, which is inclined at 30° to the horizontal. Assuming there is no friction how fast will the object be moving after it has moved down the slope a distance of 10 metres? HINT: Use an ODE that relates both distance and velocity.

Question 7.2: An object, mass m, is at rest on a plane, which is inclined at 30° to the horizontal. Assuming friction is equal to 0.1 times the contact force (i.e. the force of gravity perpendicular to the slope) how fast will the object be moving after it has slid down the slope distance of 10 metres? HINT: Alter the total forces acting on the object.

Question 7.3: An object, mass m, is at rest on a plane, which is inclined at 30° to the horizontal. Assuming friction is equal to 0.1 times the contact force (i.e. the force of gravity perpendicular to the slope) how fast will the object be moving after it has slid down the slope after 10 seconds? HINT: Alter the ODE to relate velocity and time.

Question 7.4: An object, mass m, is on on a plane, which is inclined at 30° to the horizontal. Assuming there is no friction and and an initial velocity of $2m/s$, how fast will the object be moving after it has moved down the slope a distance of 10 metres? HINT: Use an ODE that relates both distance and velocity. For this question, as you integrate the appropriate form of Newtons law, think about how to set your limits of integration. You can pair the starting speed and distance with the end speed and distance. Once you integrate with these

limits, you re-arrange to access the final distance.

Question 7.5: The drag force on a sphere which is moving through air is approximately equal to

$$\pi r^2 \frac{\rho}{2} v_{rel}^2$$

where r is the sphere's radius, v_{rel} is the velocity it is moving through the air and $\rho = 1.2$ kg m^{-3} is the density of air. A sphere, radius 0.5 m, and mass $m = 1$ kg, is pushed along a horizontal plane at an initial velocity of $v = 10\ m/s$; the wind speed is zero. How far does the sphere move before slowing down to 0.01 m/s?

Question 7.6: Derive an equation for the velocity versus distance of an object moving with an initial velocity v_0 and end velocity v_1, starting position $x = 0$ and experiencing a constant acceleration. Start by using the following form of Newtons law:

$$F = ma = mv\frac{dv}{dx}$$

noting that you can separate terms dv and dx and then integrate.

Question 7.7: Derive an equation for the velocity versus time if an object moving at an initial velocity v_0 is experiencing a constant acceleration. Start by using the following form of Newtons law:

$$F = m\frac{dv}{dt}$$

Question 7.8: A ball is thrown into the air with an initial velocity of $10m/s$. It's acceleration due to gravity (ignoring air resistance) is -9.8m/s^2. How high will the ball reach before falling back to earth?

Question 7.9: A ball is thrown into the air with an initial velocity of $10m/s$. It's acceleration is as before. How long does it take to reach it's maximum height?